Lec 2

Modeling Dynamic Behaviors

Modeling

- This part studies modeling of embedded systems, with emphasis on joint modeling of software and physical dynamics.
- The dynamics of a system is its evolution in time, i. e., how its state changes.

CPS Types

- Continuous Dynamics
- Discrete Dynamics
- Hybrid Systems

1- Continuous Dynamics

Modeling Techniques:

- Newton Mechanics
- Actor Models
- Feedback Control

Introduction

- This chapter reviews a few of the many modeling techniques for studying dynamics of a physical system.
- We begin by studying mechanical parts that move (this problem is known as classical mechanics).
- Motion of mechanical parts can often be modeled using differential equations, or equivalently, integral equations.
- Such models really only work well for "smooth" motion (a concept that we can make more precise using notions of linearity, time invariance, and continuity).

Cont.,

- For motions that are not smooth, such as those modeling collisions of mechanical parts, we can use modal models that represent distinct modes of operation with abrupt (conceptually instantaneous) transitions between modes.
- Collisions of mechanical objects can be usefully modeled as discrete, instantaneous events.

Cont.,

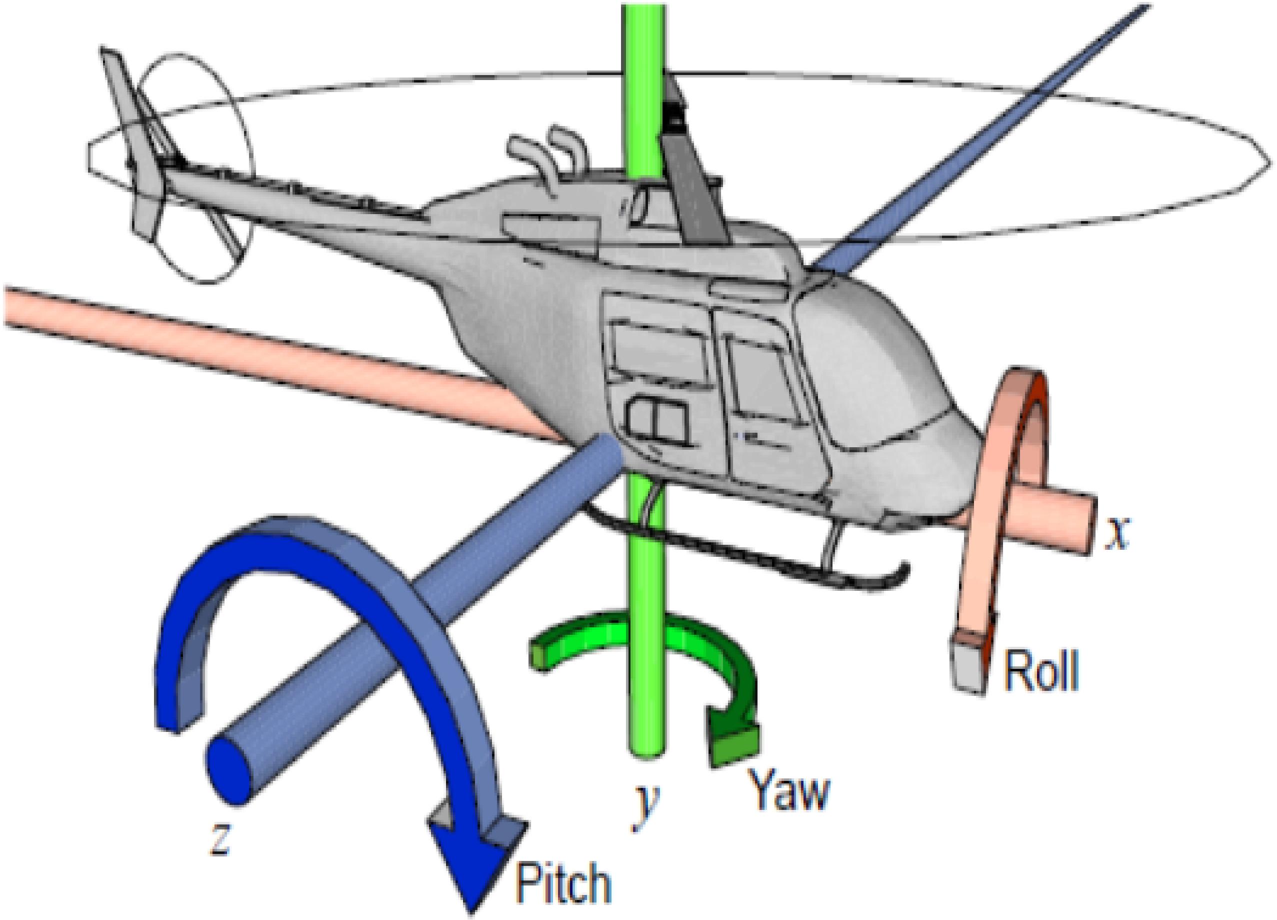
- The problem of jointly modeling smooth motion and such discrete events is known as hybrid systems modeling and is studied in Chapter 4.
- Such combinations of discrete and continuous behaviors bring us one step closer to joint modeling of cyber and physical processes.

Modeling Procedure

- We begin with simple equations of motion, which provide a model of a system in the form of ordinary differential equations (ODEs).
- We then show how these ODEs can be represented in actor models, which include the class of models in popular modeling languages such as LabVIEW (from National Instruments) and Simulink (from The MathWorks, Inc.).
- We then consider properties of such models such as linearity, time invariance, and stability, and consider consequences of these properties when manipulating models.

Newtonian Mechanics

- Motion in space of physical objects can be represented with six degrees of freedom, illustrated in the following Figure.
- Three of these represent position in three dimensional space, and three represent orientation in space: { X, Y, Z, θ_x, θ_y, θ_z }



Modeling position with six degrees of freedom requires including pitch, roll, and yaw, in addition to position.

How things fly

- **Roll**: rotation around the front to back (x axis)
- **Yaw**: rotation around the vertical axis
- **Pitch**: rotation around the side-to-side

Where in Airplane:

- Two Ailerons on the outer wings are moving in opposite direction up and down, and the plane roll to left or right.
- Elevator control pitch: on the horizontal tail surface, the elevator tilts up or down, decreasing or increasing lift on the tail. This tilts the nose of the airplane up and down.
- Rudder on the vertical tail controls Yaw. It swivels from side to side, pushing the tail to left or right directions.

Cont.,

- Roll θ_x is an angle of rotation around the x axis, where by convention an angle of 0 radians represents horizontally flat along the z axis (i.e., the angle is given relative to the z axis).
- Yaw θ_y is the rotation around the y axis, where by convention 0 radians represents pointing directly to the right (i.e., the angle is given relative to the x axis).
- Pitch θ_z is rotation around the z axis, where by convention 0 radians represents pointing horizontally (i.e., the angle is given relative to the x axis).

Cont.,

- The position of an object in space, therefore, is represented by six functions of the form f : R → R, where the domain represents time and the codomain represents either distance along an axis or angle relative to an axis.
- Changes in position or orientation are governed by Newton’s second law, relating force with acceleration.

$$\mathbf{F}(t) = M\ddot{\mathbf{x}}(t), \qquad (2.1)$$

- Where, F is the force vector in 3-D, M is the mass of the object, and $X^{\cdot\cdot}$ is the acceleration.

Velocity & Position

- Velocity is the integral of acceleration.

$$\forall\, t > 0, \quad \dot{\mathbf{x}}(t) \;=\; \dot{\mathbf{x}}(0) + \int_0^t \ddot{\mathbf{x}}(\tau)d\tau$$

where $\dot{\mathbf{x}}(0)$ is the initial velocity in three directions. Using (2.1), this becomes

$$\forall\, t > 0, \quad \dot{\mathbf{x}}(t) = \dot{\mathbf{x}}(0) + \frac{1}{M}\int_0^t \mathbf{F}(\tau)d\tau,$$

Position is the integral of velocity,

$$\begin{aligned}\mathbf{x}(t) &= \mathbf{x}(0) + \int_0^t \dot{\mathbf{x}}(\tau)d\tau \\ &= \mathbf{x}(0) + t\dot{\mathbf{x}}(0) + \frac{1}{M}\int_0^t \int_0^\tau \mathbf{F}(\alpha)d\alpha d\tau,\end{aligned}$$

Position

- Using these equations, if you know the initial position and initial velocity of an object and the forces on the object in all three directions as a function of time, you can determine the acceleration, velocity, and position of the object at any time.

Orientation

- By using Torque (N/m), the rotational version of force around an axis.
- It is again a three-element vector as a function of time, representing the net rotational force on an object.
- It can be related to angular velocity.

$$\mathbf{T}(t) = \frac{d}{dt}\left(\mathbf{I}(t)\dot{\theta}(t)\right),$$

- where T is the torque vector in three axes and I(t) is the moment of inertia tensor of the object in 3-D.

To be explicit about the three dimensions, we might write (2.2) as

$$\begin{bmatrix} T_x(t) \\ T_y(t) \\ T_z(t) \end{bmatrix} = \frac{d}{dt}\left(\begin{bmatrix} I_{xx}(t) & I_{xy}(t) & I_{xz}(t) \\ I_{yx}(t) & I_{yy}(t) & I_{yz}(t) \\ I_{zx}(t) & I_{zy}(t) & I_{zz}(t) \end{bmatrix}\begin{bmatrix} \dot{\theta}_x(t) \\ \dot{\theta}_y(t) \\ \dot{\theta}_z(t) \end{bmatrix}\right).$$

Here, for example, $T_y(t)$ is the net torque around the y axis (which would cause changes in yaw), $I_{yx}(t)$ is the inertia that determines how acceleration around the x axis is related to torque around the y axis.

Cont.,

- If the object is spherical, for example, this reluctance is the same around all axes, so it reduces to a constant scalar I (or equivalently, to a diagonal matrix I with equal diagonal elements I).
- The equation then looks much more like :

$$\mathbf{T}(t) = I\ddot{\theta}(t).$$

Rotational velocity is the integral of acceleration,

$$\dot{\theta}(t) = \dot{\theta}(0) + \int_0^t \ddot{\theta}(\tau)d\tau,$$

where $\dot{\theta}(0)$ is the initial rotational velocity in three axes. For a spherical object, using (2.3), this becomes

$$\dot{\theta}(t) = \dot{\theta}(0) + \frac{1}{I}\int_0^t \mathbf{T}(\tau)d\tau.$$

Orientation is the integral of rotational velocity,

$$\begin{aligned}
\theta(t) &= \theta(0) + \int_0^t \dot{\theta}(\tau)d\tau \\
&= \theta(0) + t\dot{\theta}(0) + \frac{1}{I}\int_0^t \int_0^\tau \mathbf{T}(\alpha)d\alpha d\tau
\end{aligned}$$

Example

- A helicopter has two rotors, one above, which provides lift, and one on the tail counteracts spinning.
- The force produced by the tail rotor must counter the torque produced by the main rotor.

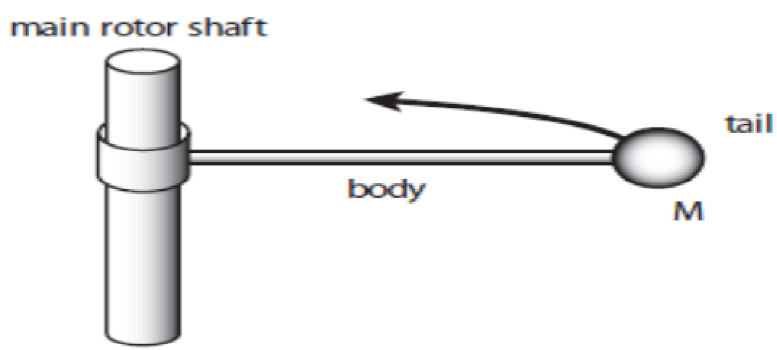


Figure 2.2: Simplified model of a helicopter.

Modeling Physical Dynamics: Feedback Control Problem

A helicopter without a tail rotor, like the one below, will spin uncontrollably due to the torque induced by friction in the rotor shaft.

Control system problem:
Apply torque using the tail rotor to counterbalance the torque of the top rotor.

Cont.,

- Assume that the helicopter position is fixed at the origin.
- Assume that the helicopter remains vertical, so pitch and roll are fixed at zero.
- The moment of Inertia (I) reduces to a scalar quantity that resist the changes in Yaw.
- The changes in yaw will be due to Newton's third law, the action-reaction law, which states that every action has an equal and opposite reaction.

Cont.,

- We can model the simplified helicopter by a system that takes as input a continuous time signal Ty, the torque around the y axis (which causes changes in yaw).
- This torque is the sum of the torque caused by the main rotor and that caused by the tail rotor.
- When these are perfectly balanced, that sum is zero.
- The output of our system will be the angular velocity around the y axis:

$$\dot{\theta}_y(t) = \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau.$$

Actor Model of Systems

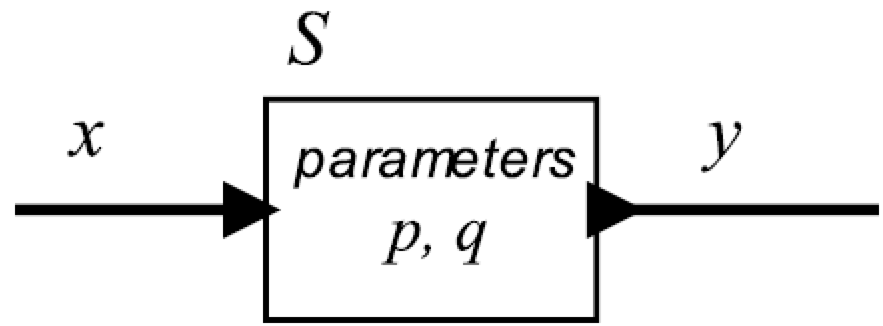
Is a SW graphical model mirrors for ODE.

It is a box, where the inputs are function, and outputs also are functions.

A *system* is a function that accepts an input *signal* and yields an output signal.

The domain represents the time, and the codomain represents the value of the signal at a particular time.

Parameters may affect the definition of the function *S*.



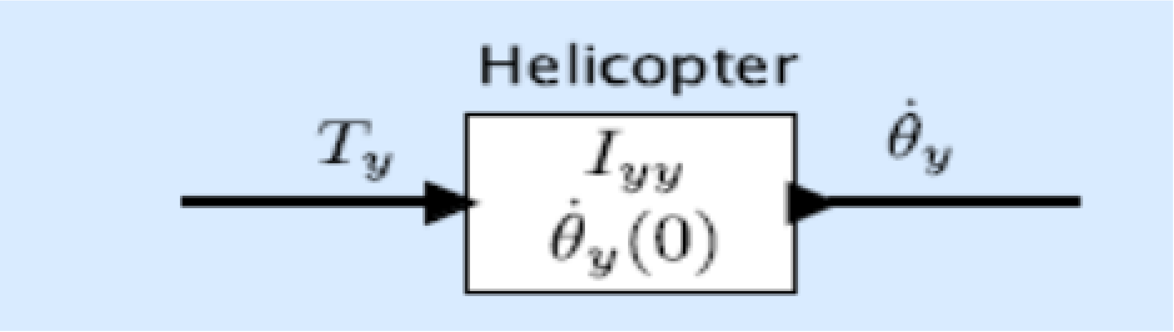
$$x:\mathbb{R} \to \mathbb{R}, \qquad y:\mathbb{R} \to \mathbb{R}$$

$$S:X \to Y$$

$$X = Y = (\mathbb{R} \to \mathbb{R})$$

Example

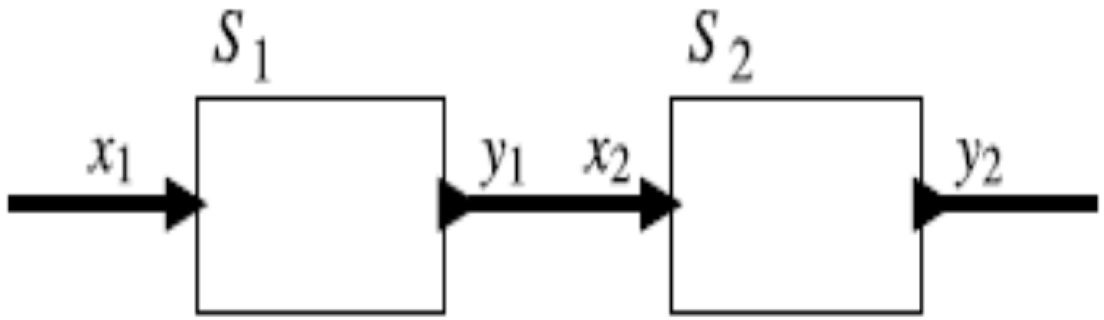
- The actor model for the helicopter of the above example can be depicted as follows:



Characteristics of Actor Model

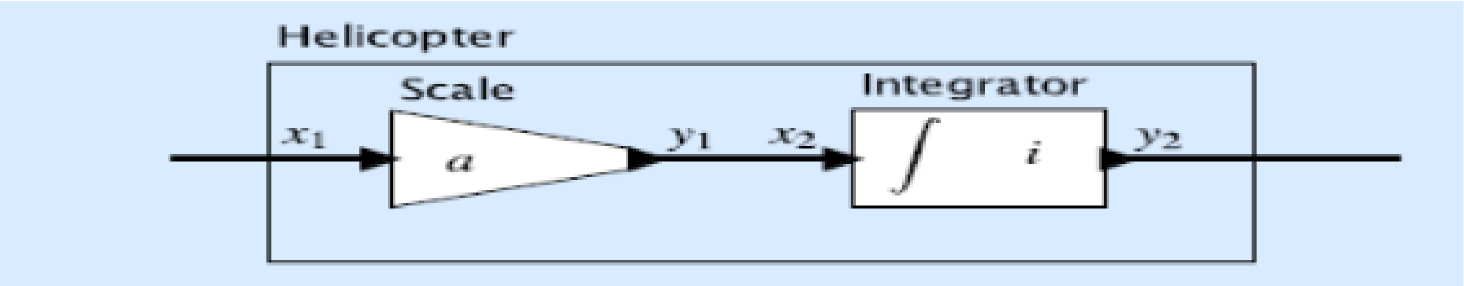
1- Composable.

- We can form a cascaded composition



Example

- The actor model for the helicopter can be represented as a cascade composition of two actors as follows:

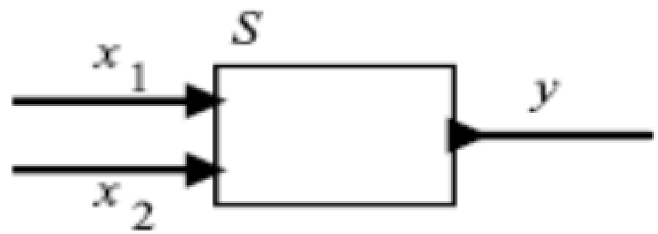


$$\forall\, t \in \mathbb{R}, \quad y_2(t) = i + \int_0^t x_2(\tau) d\tau.$$

If we give the parameter values $a = 1/I_{yy}$ and $i = \dot{\theta}_y(0)$, we see that this system represents (2.4) where the input $x_1 = T_y$ is torque and the output $y_2 = \dot{\theta}_y$ is angular velocity.

Cont.,

2- Multiple input signals and/or multiple output signals.



$$\forall\, t \in \mathbb{R}, \quad y(t) = x_1(t) + x_2(t).$$

Properties of Systems:

1- Causal Systems

A system is causal if its output depends only on current and past inputs.

2- **Memoryless Systems**

if the output depends only on the current inputs.

3- **Linearity and Time Invariance**

Is a system that is both linear and time invariant.

4- **Stability**

If the output signal is bounded for all input signals that are bounded.

Example

- In the helicopter system developed in the above Example.
- Let the input be Ty = u, where u is the unit step, given by:

$$\forall\, t \in \mathbb{R}, \quad u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}.$$

- This means that prior to time zero, there is no torque applied to the system, and starting at time zero, we apply a torque of unit magnitude. This input is clearly bounded.
- However, the output velocity grows without bound.
- In practice, a helicopter uses a feedback system to determine how much torque to apply at the tail rotor to keep the body of the helicopter straight.

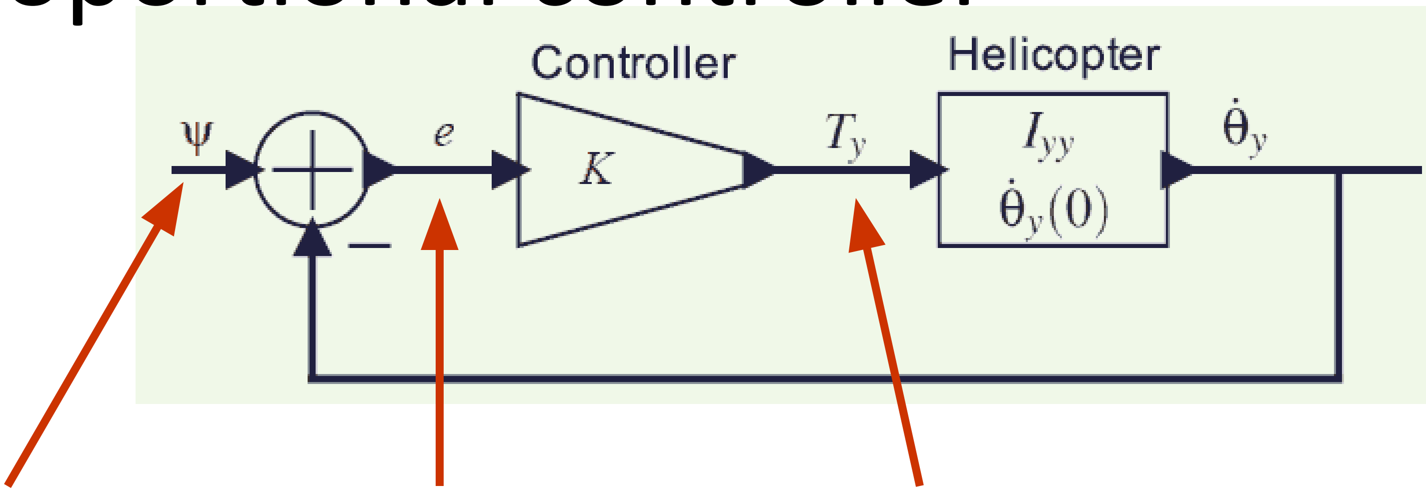
Feedback Control

- A system with feedback has directed cycles, where an output from an actor is fed back to affect an input of the same actor.
- Most control systems use feedback. They make measurements of an error (e), which is a discrepancy between desired behavior and actual behavior and use that measurement to correct the behavior.

Example 2-5

- Recall that the helicopter model of Example 2.1 is not stable.
- We can stabilize it with a simple feedback control system, as shown in the previous Figure
- assume that $\psi(t) = 0$ for all t; i.e., we wish to control the helicopter simply to keep it from rotating at all.
- The desired angular velocity is zero.

Proportional controller



desired

angular

velocity

error

signal

net

torque

$$e(t) = \psi(t) - \dot{\theta}_y(t) \qquad T_y(t) = Ke(t)$$

$$\begin{aligned}
\dot{\theta}_y(t) &= \dot{\theta}_y(0) + \frac{1}{I_{yy}} \int_0^t T_y(\tau)\,d\tau \\
&= \dot{\theta}_y(0) + \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau))\,d\tau
\end{aligned}$$

$$\dot{\theta}_y(t) = \dot{\theta}_y(0)e^{-Kt/I_{yy}}u(t).$$

For larger K, it will approach more quickly. For negative K, the system is unstable, and angular velocity will grow without bound

Example 2.6:

- Assume that helicopter is initially at rest.

$$\dot{\theta}(0) = 0,$$

and that the desired signal is

$$\psi(t) = au(t)$$

for some constant a. That is, we wish to control the helicopter to get it to rotate at a fixed rate.

We use (2.4) to write

$$\begin{aligned}
\dot{\theta}_y(t) &= \frac{1}{I_{yy}} \int_0^t T_y(\tau) d\tau \\
&= \frac{K}{I_{yy}} \int_0^t (\psi(\tau) - \dot{\theta}_y(\tau)) d\tau \\
&= \frac{K}{I_{yy}} \int_0^t a d\tau - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau \\
&= \frac{Kat}{I_{yy}} - \frac{K}{I_{yy}} \int_0^t \dot{\theta}_y(\tau) d\tau.
\end{aligned}$$

Using the same (black magic) technique of inferring and then verifying the solution, we can see that the solution is

$$\dot{\theta}_y(t) = au(t)(1 - e^{-Kt/I_{yy}}). \tag{2.13}$$

Note:

- The above example is somewhat unrealistic because we cannot independently control the net torque of the helicopter.
- In particular, the net torque Ty is the sum of the torque Tt due to the top rotor and the torque Tr due to the tail rotor.

$$\forall\, t \in \mathbb{R}, \quad T_y(t) = T_t(t) + T_r(t)\ .$$

- Thus, we will actually need to design a control system that controls Tr and stabilizes the helicopter for any Tt.

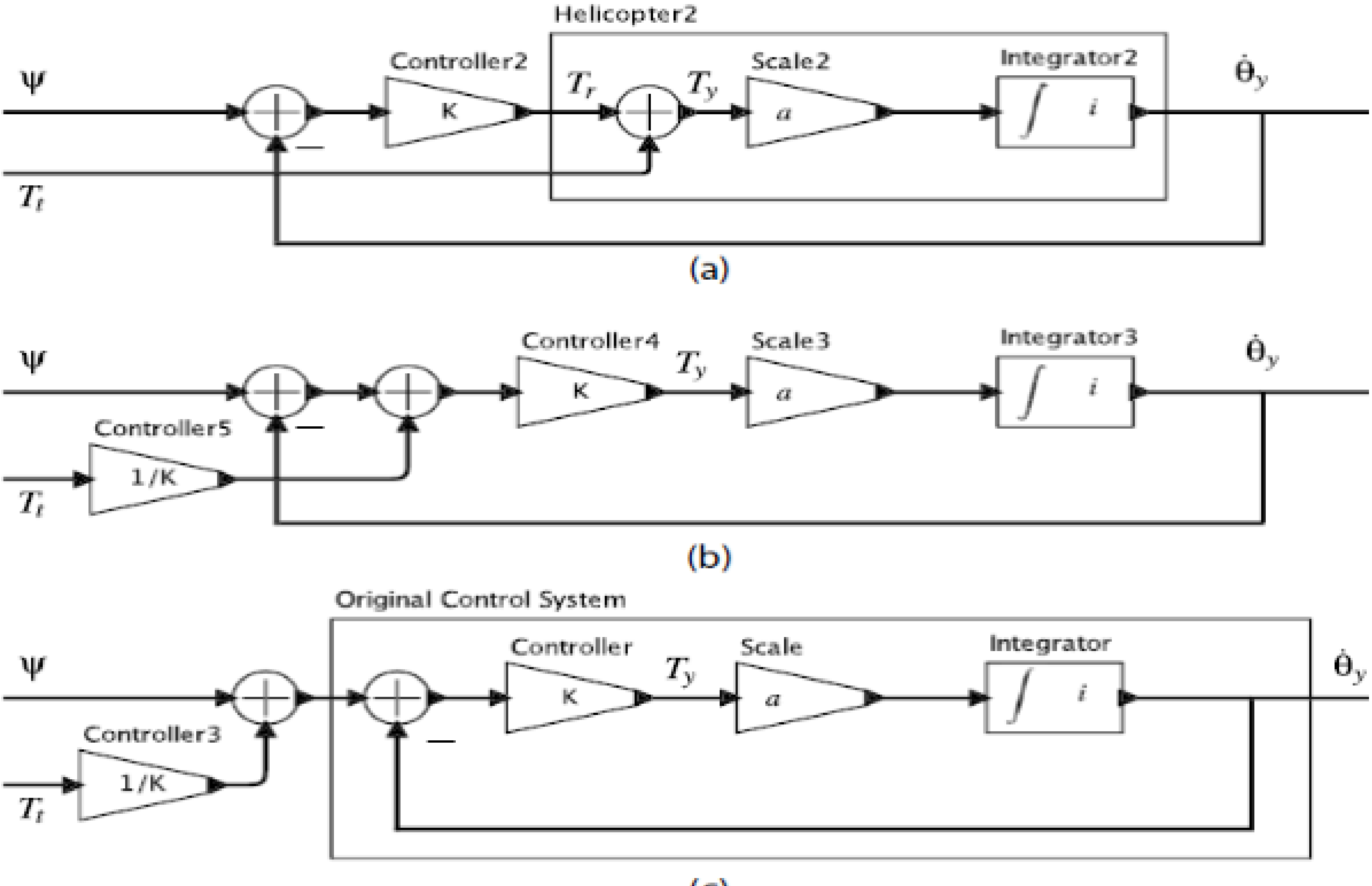
Example 2.7:

- We have to modify the helicopter model so that it has two inputs, Tt and Tr.
- The feedback control system is now controlling only Tr, and Tt is treated as an external (uncontrolled) input signal.
- How well will this control system behave?
- Suppose that the torque due to the top rotor is given by:

$$T_t = bu(t)$$

Cont.,

- That is, at time zero, the top rotor starts spinning a constant velocity, and then holds that velocity.
- Suppose further that the helicopter is initially at rest.
- Find the behavior of the system.


Helicopter2
Controller2
K
Scale2
a
Integrator2
i
ψ
T_r
T_y
$\dot{\theta}_y$
T_t
(a)
Controller4
K
Scale3
a
Integrator3
i
Controller5
1/K
ψ
T_y
$\dot{\theta}_y$
T_t
(b)
Original Control System
Controller
K
Scale
a
Integrator
i
Controller3
1/K
ψ
T_y
$\dot{\theta}_y$
T_t
(c)

Sheet #1

- Examples 2-5, 2-6, and 2-7 using MATLAB.